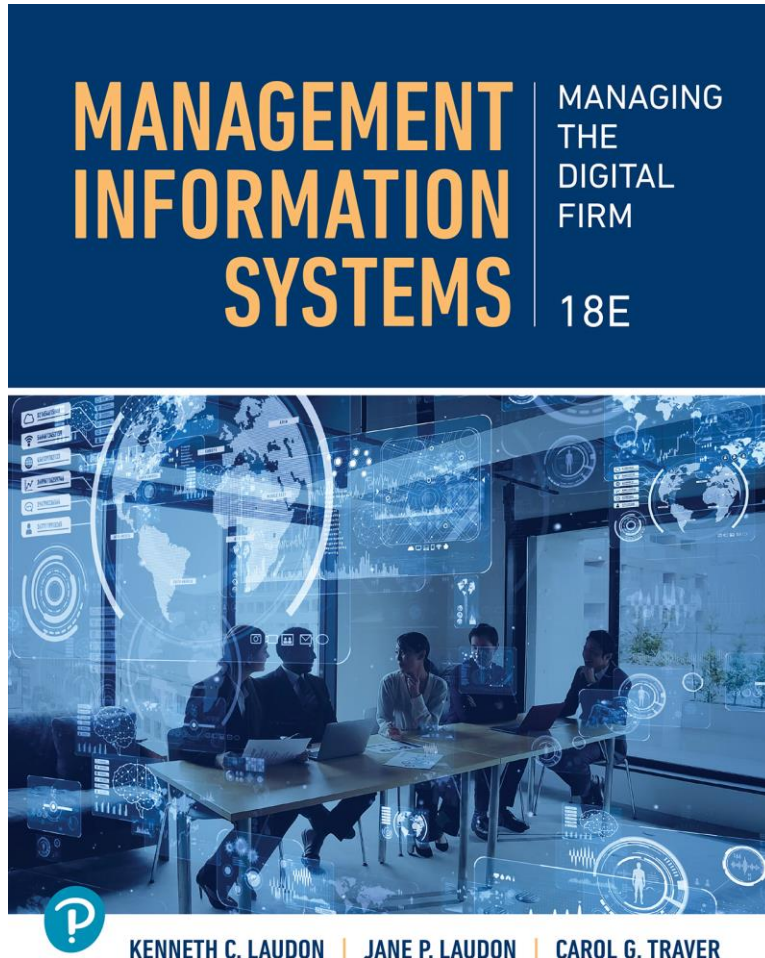


# Management Information Systems: Managing the Digital Firm

Eighteenth Edition



## Chapter 9

**Achieving Operational  
Excellence and Customer  
Intimacy: Enterprise**

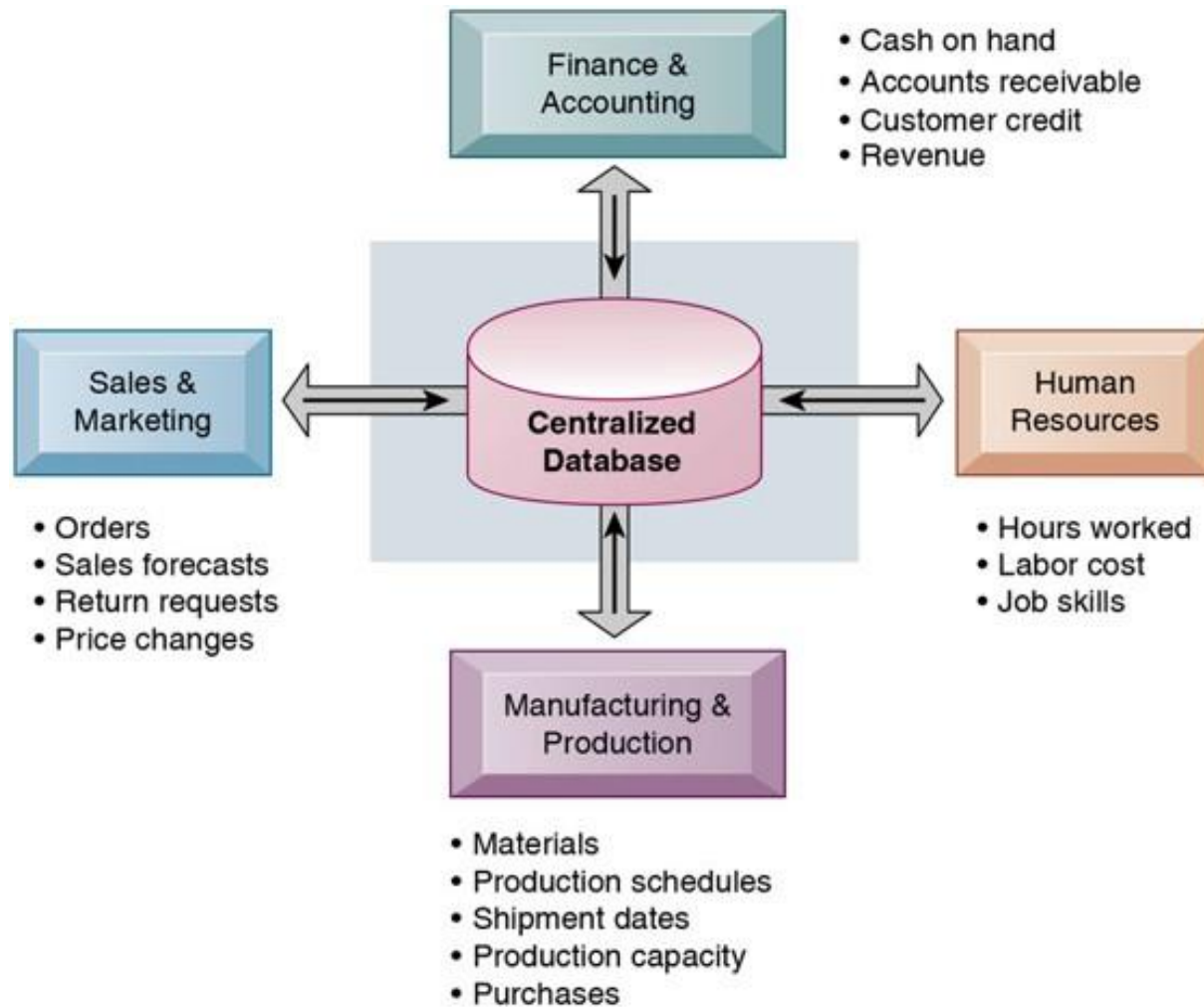
# Learning Objectives

- 9.1 Understand ERP systems.
- 9.2 Understand SCM systems.
- 9.3 Describe global supply chain management challenges.
- 9.4 Understand CRM systems.
- 9.5 Discuss enterprise application challenges.
- 9.6 Describe how enterprise applications are using AI.
- 9.7 Understand how the information in this chapter can help your career.

# ERP Systems (1 of 2)

- What is an ERP System?
  - Enterprise Resource Planning (ERP) systems
    - Based on a suite of integrated software modules and a common central database
    - Collects data from many divisions of firm for use in nearly all of firm's internal business activities
    - Information entered in one process is immediately available for other processes

# Figure 9.1 How ERP Systems Work



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## Table 9.1

### Business Processes Supported by ERP Systems

**Financial and accounting processes**, including general ledger, accounts payable, accounts receivable, fixed assets, cash management and forecasting, product-cost accounting, cost center accounting, asset accounting, tax accounting, credit management, and financial reporting

**Human resources processes**, including personnel administration, time accounting, payroll, personnel planning and development, benefits accounting, applicant tracking, time management, compensation, workforce planning, performance management, and travel expense reporting

**Manufacturing and production processes**, including procurement, inventory management, purchasing, shipping, production planning, production scheduling, material requirements planning, quality control, distribution, transportation execution, and plant and equipment maintenance

**Sales and marketing processes**, including order processing, quotes, contracts, product configuration, pricing, billing, credit checking, incentive and commission management, and sales planning

# ERP Systems (2 of 2)

- Built around thousands of predefined business processes that reflect best practices
  - Finance and accounting
  - Human resources
  - Manufacturing and production
  - Sales and marketing
- To implement, firms:
  - Select functions of system they wish to use
  - Map business processes to software processes
    - Use software's configuration tables for customizing

# Business Value of ERP Systems

- Increased operational efficiency
- Provide firm-wide information to support decision making
- Enable rapid responses to customer requests for information or products
- Include analytical tools to evaluate overall organizational performance and improve decision-making

# AI and ERP Systems

- ERP system vendors are increasingly incorporating AI
- Many ERP systems now include natural language processing capabilities
- AI-driven ERP systems can identify patterns in very large volumes of data

**Table 9.2**

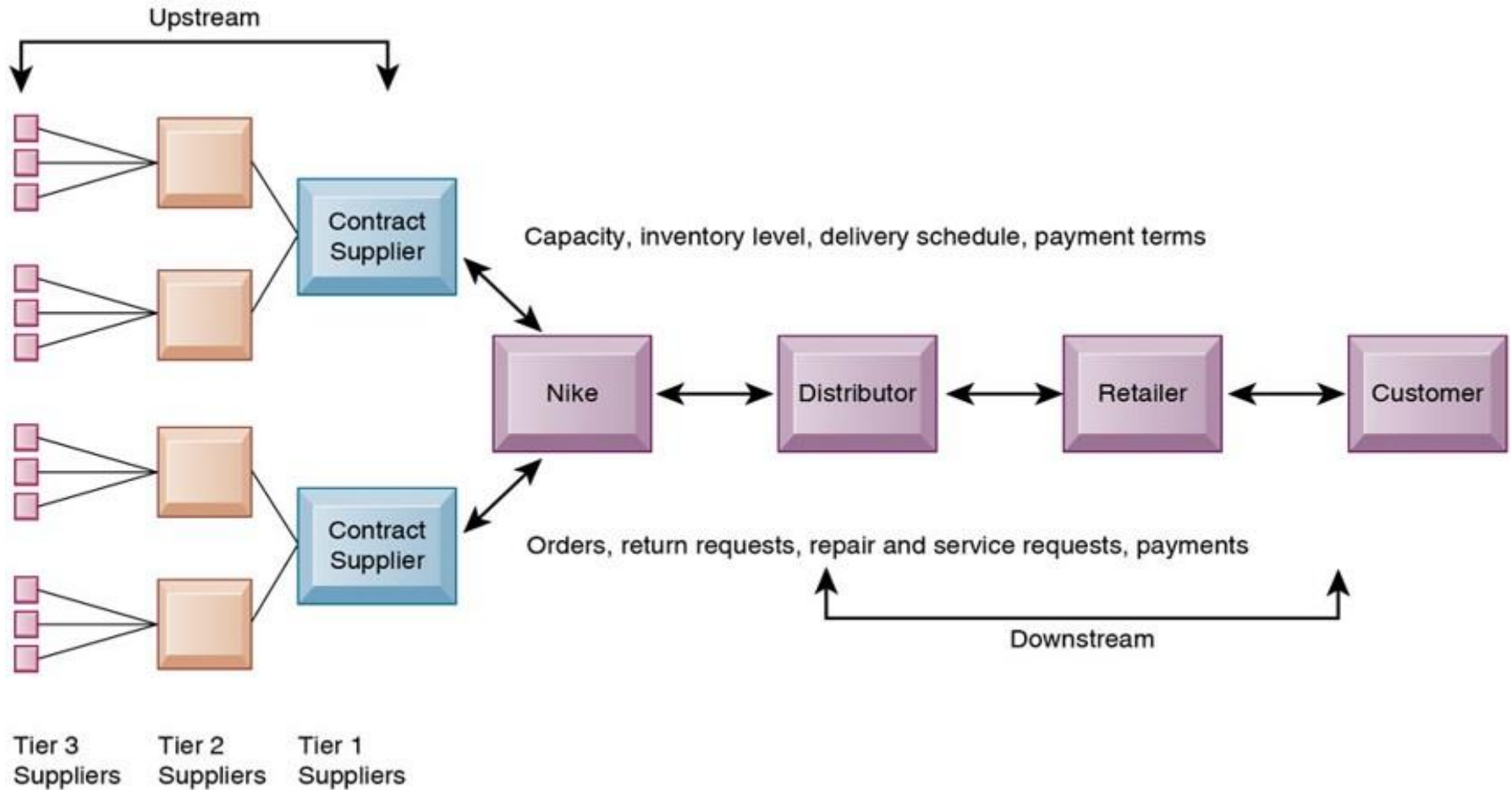
Examples of AI Applications in ERP Systems

| Capability                                 | Description                                                                                                                                                                                        |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Digital assistant                          | Use AI-powered conversational chatbots to respond to customer and employee inquiries by providing support, answering questions, capturing details, and resolving issues without human interaction. |
| Predictive analytics                       | Anticipating equipment failures, recommending preventative maintenance schedules and procedures                                                                                                    |
| Anomaly detection                          | Analyzing ERP financial reports to identify irregularities in data entries or unusual transactions                                                                                                 |
| Improve product development and innovation | Suggest new product ideas, optimize product features and assortment                                                                                                                                |
| Financial forecasting                      | Generating plausible financial scenarios based on potential economic, policy, or market changes                                                                                                    |
| Optimize workflows                         | AI can assist with almost every function of a modern supply chain, including planning, inventory and warehouse management, transaction processing, transportation, monitoring, and inspection.     |

# Supply Chain Management (SCM) Systems

- The Supply Chain
  - Network of organizations and processes for:
    - Procuring materials
    - Transforming materials into products
    - Distributing the products
  - Upstream supply chain
  - Downstream supply chain
  - Internal supply chain

# Figure 9.2 Nike's Supply Chain

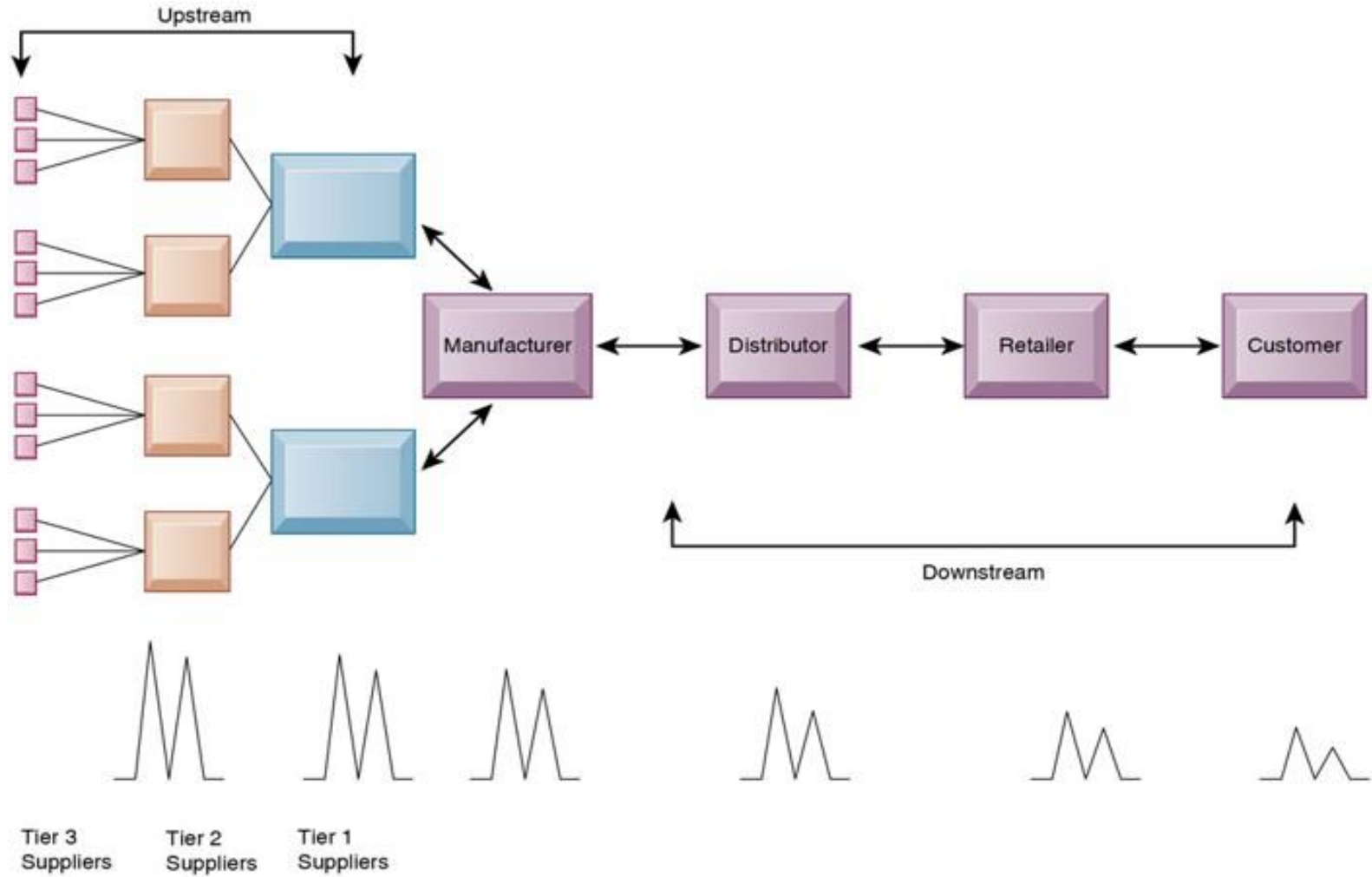


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# Information Systems and SCM

- Inefficiencies cut into a company's operating costs
  - Can waste up to 25 percent of operating expenses
- **Just-in-time strategy**
  - Components arrive as they are needed
  - Finished goods shipped after leaving assembly line
- **Safety stock:** buffer for lack of flexibility in supply chain
- **Bullwhip effect**
  - Information about product demand gets distorted as it passes from one entity to next across supply chain

# Figure 9.3 The Bullwhip Effect



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# SCM Software

- **Supply chain planning systems**

- Model existing supply chain
- Enable demand planning
- Optimize sourcing, manufacturing plans
- Establish inventory levels
- Identify transportation modes

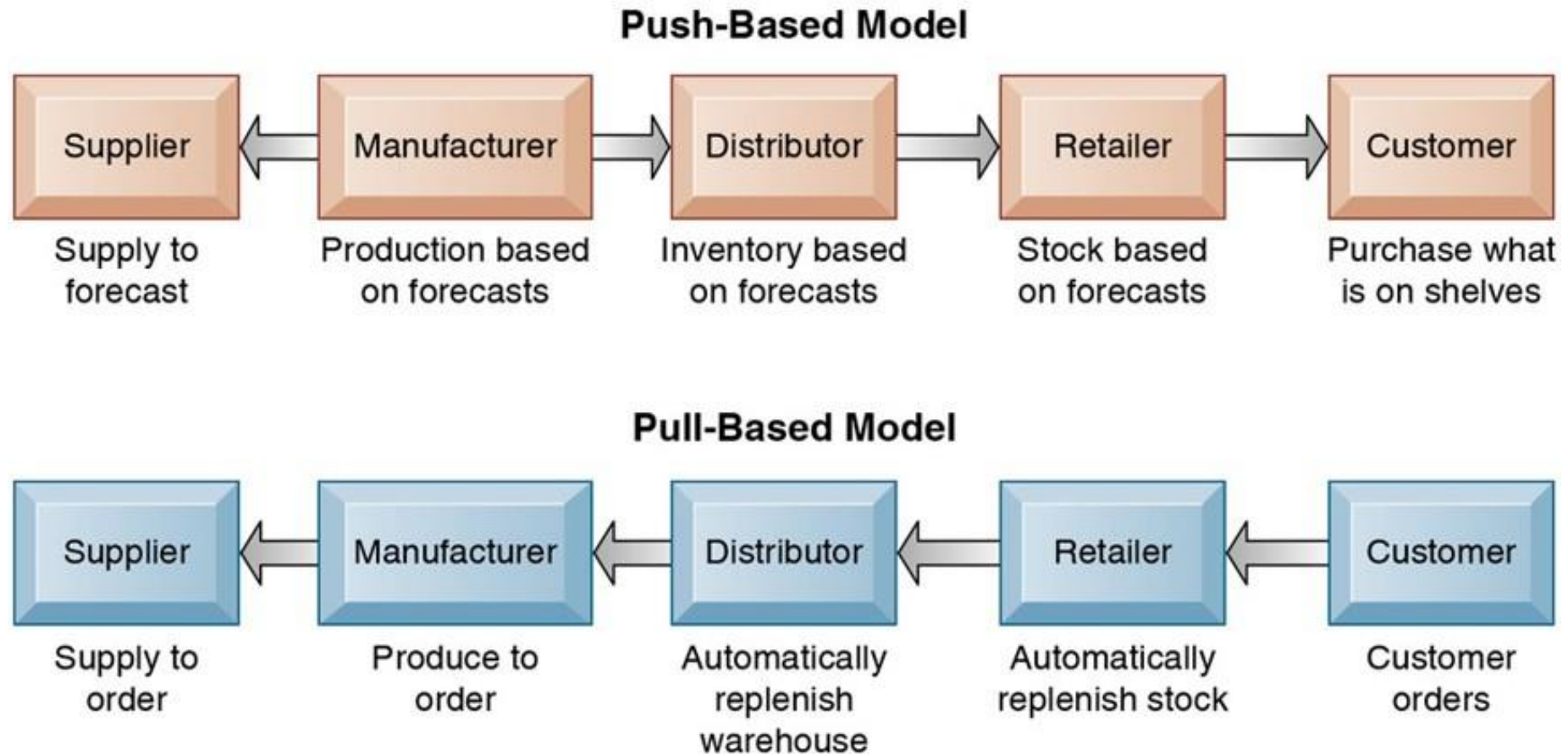
- **Supply chain execution systems**

- Manage flow of products through distribution centers and warehouses

# Demand-Driven Supply Chains

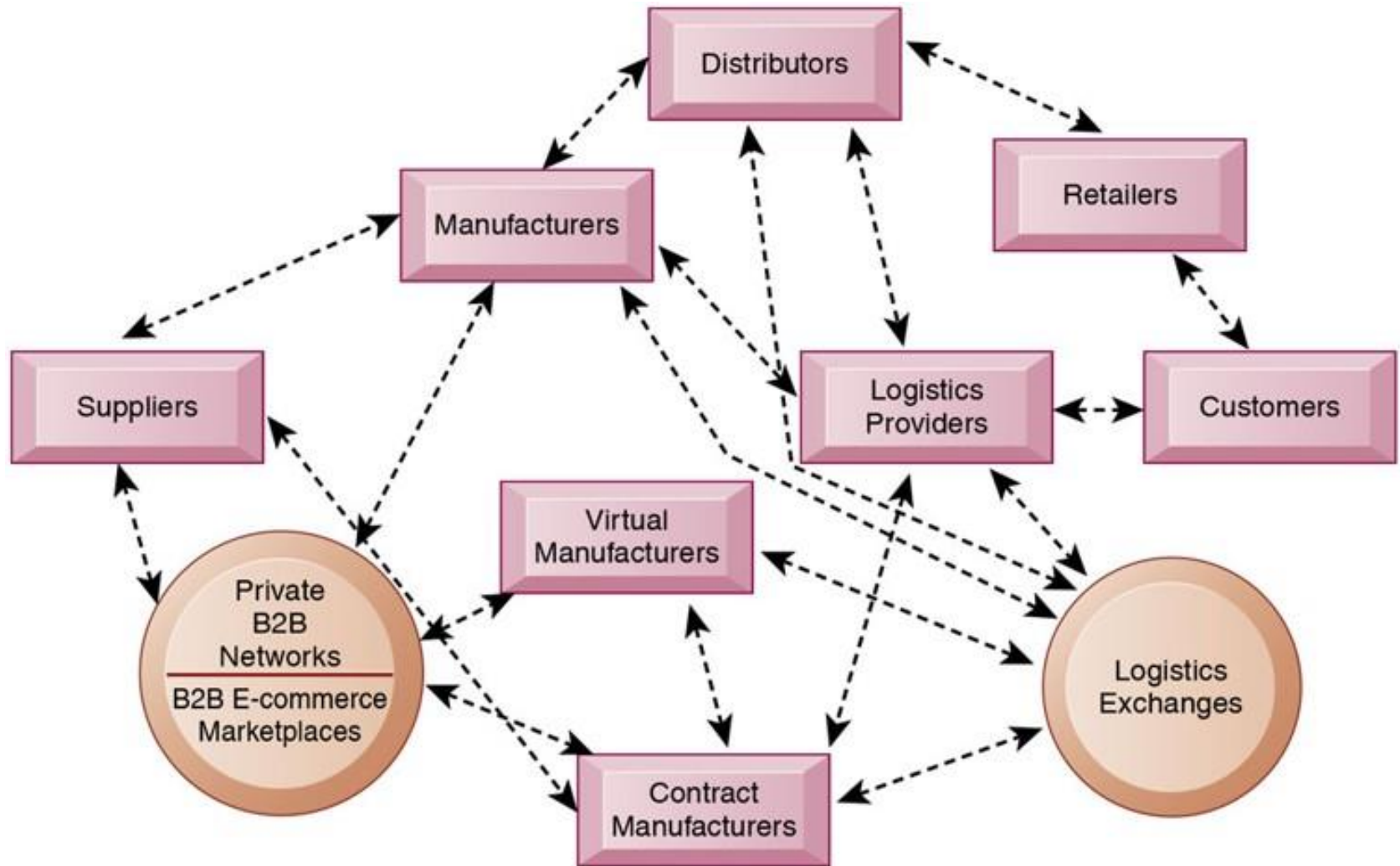
- **Push-based model (build-to-stock)**
  - Driven by forecasts or best guesses of demand for products, rather than what the customer orders—“build-to-stock”
- **Pull-based model (demand-driven)**
  - Customer orders trigger events in supply chain
- Internet enables move from sequential supply chains to concurrent supply chains
  - Complex networks of suppliers can adjust immediately

# Figure 9.4 Push vs. Pull-Based Supply Chain Models



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# Figure 9.5 The Internet-Driven Supply Chain



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# Business Value of SCM Systems

- Match supply to demand
- Reduce inventory levels
- Improve delivery service
- Speed product time to market
- Use assets more effectively
  - Total supply chain costs can be 75 percent of operating budget
- Increase sales

# Global SCM Challenges

- Global supply chain issues
  - Geographical distances, time differences
  - Participants from different countries
    - Different performance standards
    - Different legal requirements
- Internet helps manage global complexities
  - Warehouse management
  - Transportation management
  - Logistics
  - Outsourcing

# AI and Supply Chain Management

- AI is very useful for analyzing the proliferating amount of big data generated by modern global supply chains
  - To develop more accurate forecasts
  - Reveal operational insights
  - Improve efficiency of storage and transportation processes across vast logistics networks with multiple partners

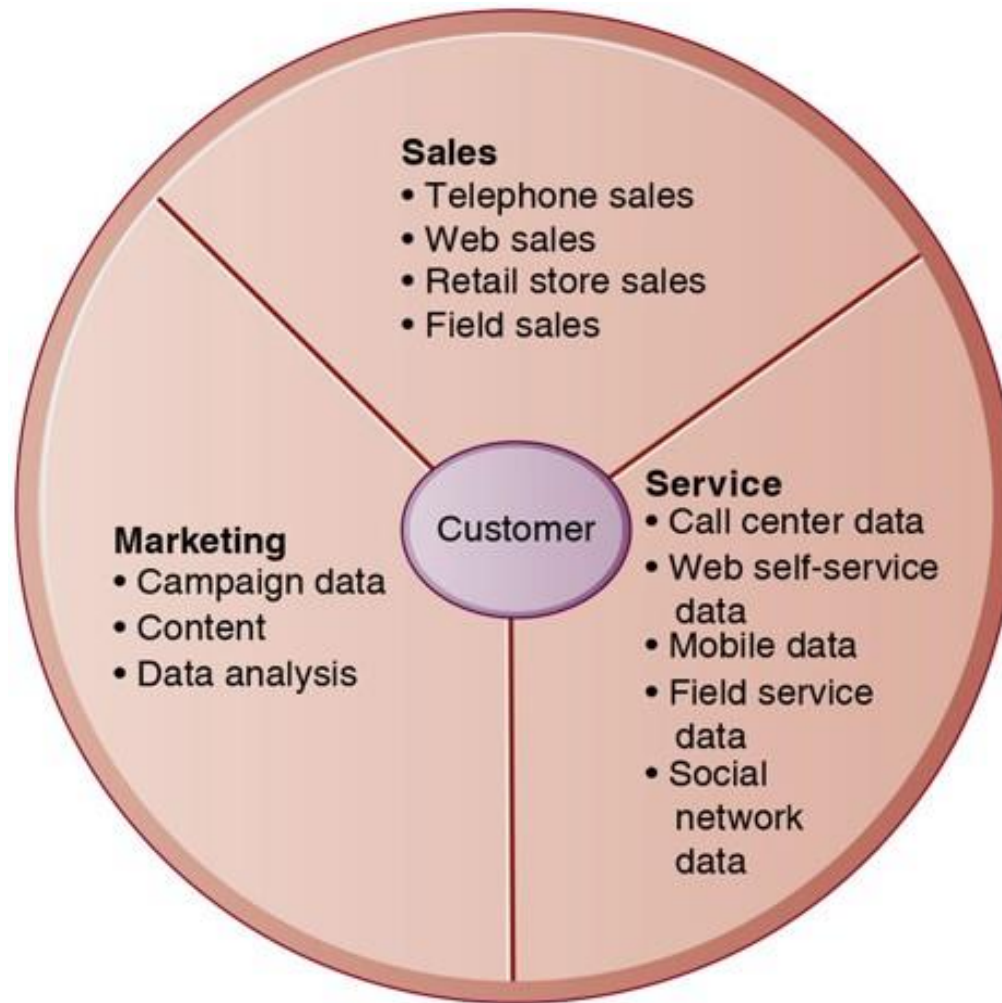


| Capability                                | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| More accurate inventory management        | AI can manage inventory levels with greater precision and efficiency. AI-powered forecasting systems can use inventory information to forecast customer demand and adjust product stocking accordingly. Computer vision systems (such as cameras installed on supply chain infrastructure, vehicles, and drones) can calculate inventory levels in real time throughout the supply chain and monitor warehouse storage capacity.                                                                                                                                                          |
| More timely deliveries                    | Logistics companies use machine learning to train models that optimize and manage the delivery routes as components move along the supply chain. These models can prioritize shipments based on order volumes, delivery promises, contractual deadlines, customer importance, or product availability. AI can also provide all parties in distribution networks with more-accurate estimated times of arrival.                                                                                                                                                                            |
| Improved supply chain sustainability      | AI can drive operational efficiencies that make supply chains more sustainable, such as providing information to help organizations reduce delivery vehicle fuel consumption or wasted materials at various steps in the supply chain. More precise demand forecasts and production planning help ensure that manufacturers don't overproduce.                                                                                                                                                                                                                                            |
| More precise demand planning              | AI has become the gold standard for predicting demand based on both internal signals, such as sales patterns, marketing leads, economic outlook, and broader market trends. AI-infused demand planning software is capable of estimating the potential impact of external phenomena such as severe weather events, economic downturns, and political disruptions.                                                                                                                                                                                                                         |
| Simulations for supply chain optimization | AI-powered simulations provide visual insights into the operations of complex logistical networks and recognize ways to improve them. When AI selects the models and controls the workflows, these simulations become more accurate than those run with traditional computing methods. This application of AI can help engineers and production managers assess the impacts of redesigning products. Planners and operations managers can evaluate the potential impact of changing suppliers, redirecting shipping and distribution routes, or relocating storage and distribution hubs. |

# Customer Relationship Management (CRM) Systems (1 of 3)

- Customer relationship management (CRM) system
  - Captures and integrates customer data from all over the organization
  - Consolidates and analyzes customer data
  - Distributes customer information to various systems and customer touch points across enterprise
  - Provides a single enterprise view of customers

# Figure 9.6 CRM Systems



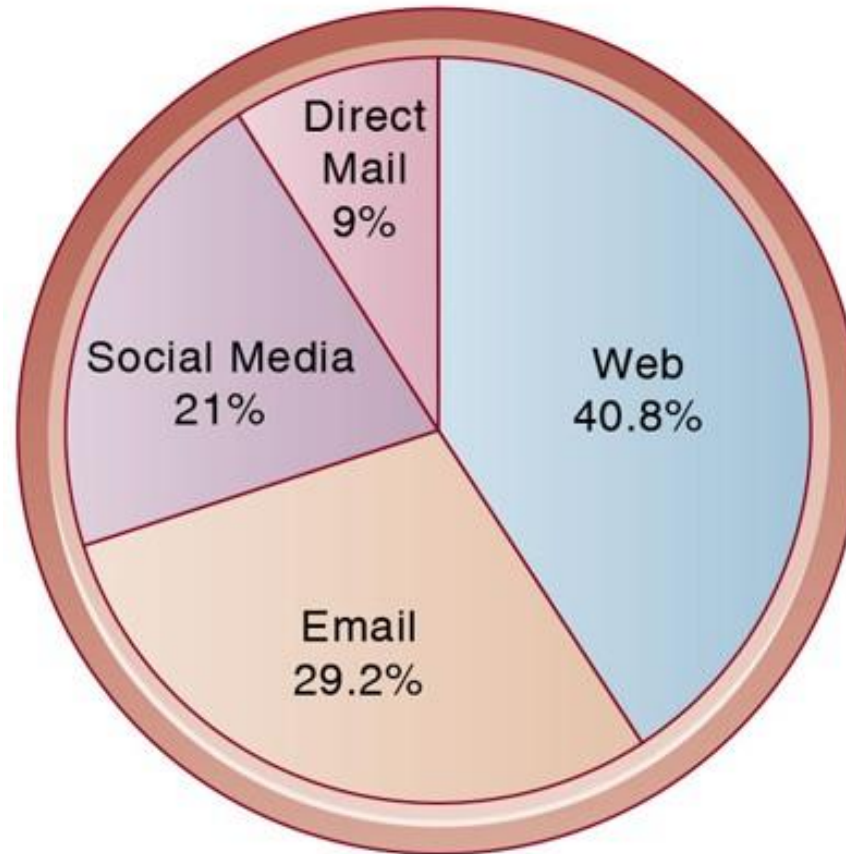
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# CRM Systems (2 of 3)

- Commercial CRM system software packages range from niche tools to large-scale enterprise applications
  - Sales force automation (SFA modules)
    - Sales prospect and contact information
    - Sales quote generation capabilities
  - Customer service
    - Assigning and managing customer service requests
    - Web-based self-service capabilities
  - Marketing
    - Capturing prospect and customer data, scheduling and tracking direct-marketing mailings or e-mail
    - Cross-selling

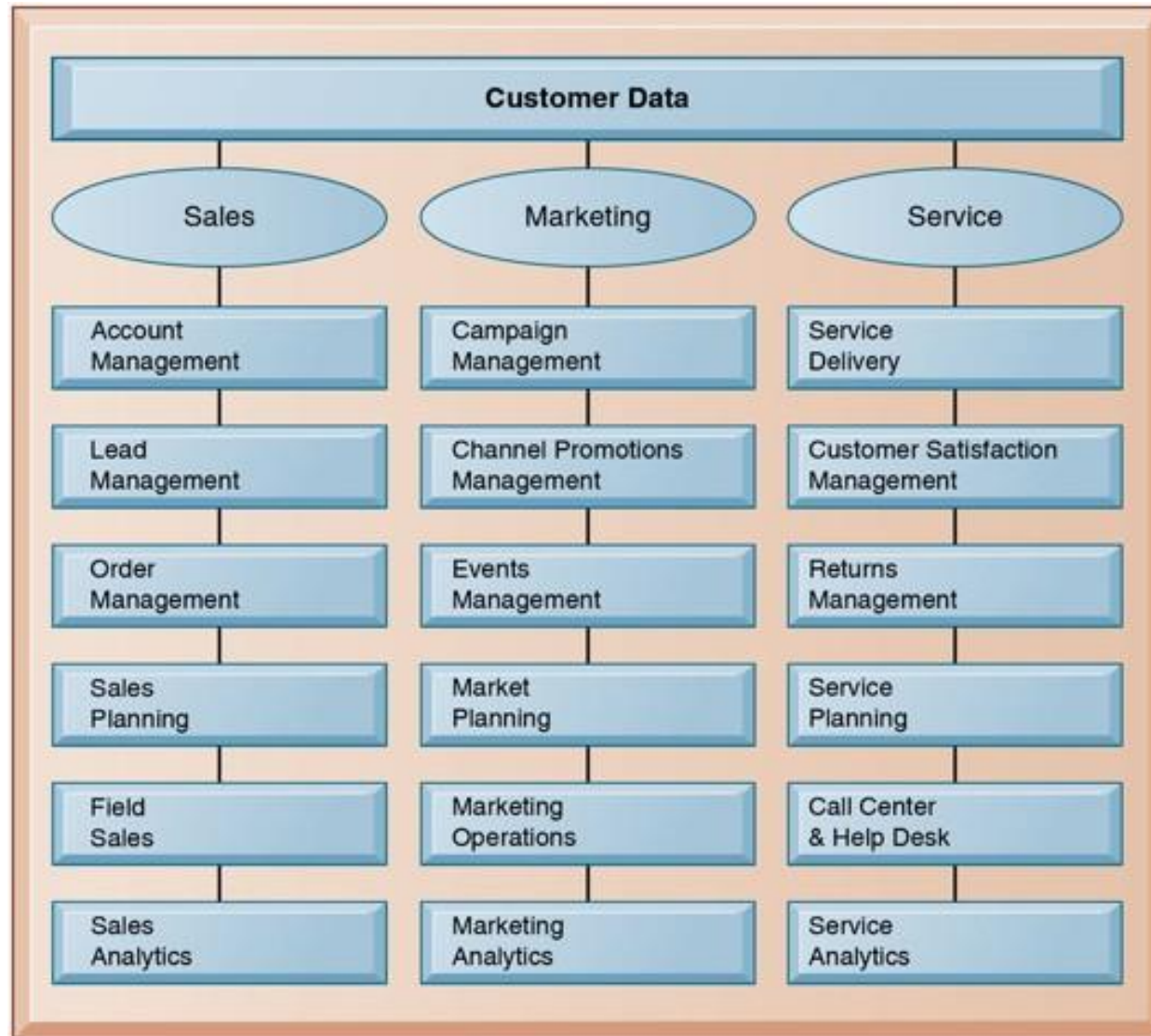
# Figure 9.7 How CRM Systems Support Marketing

**Responses by Channel for January 2024  
Promotional Campaign**



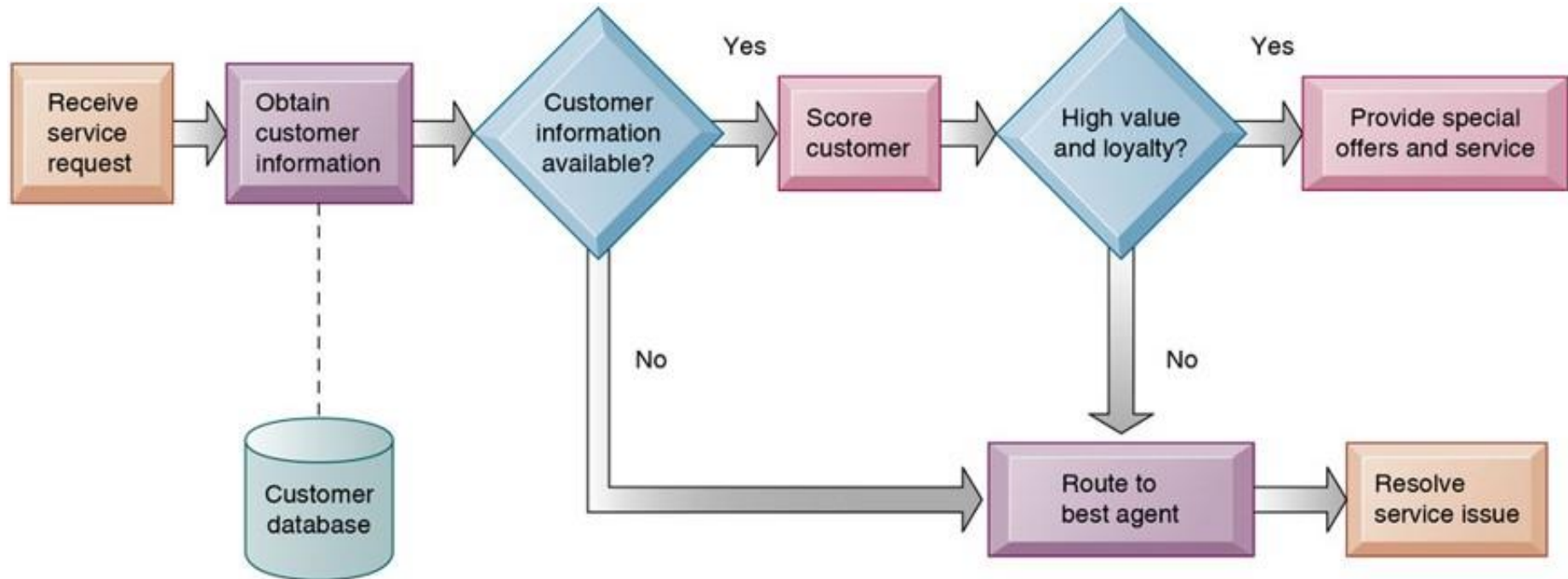
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# Figure 9.8 CRM System Capabilities



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# Figure 9.9 Customer Loyalty Management Process Map

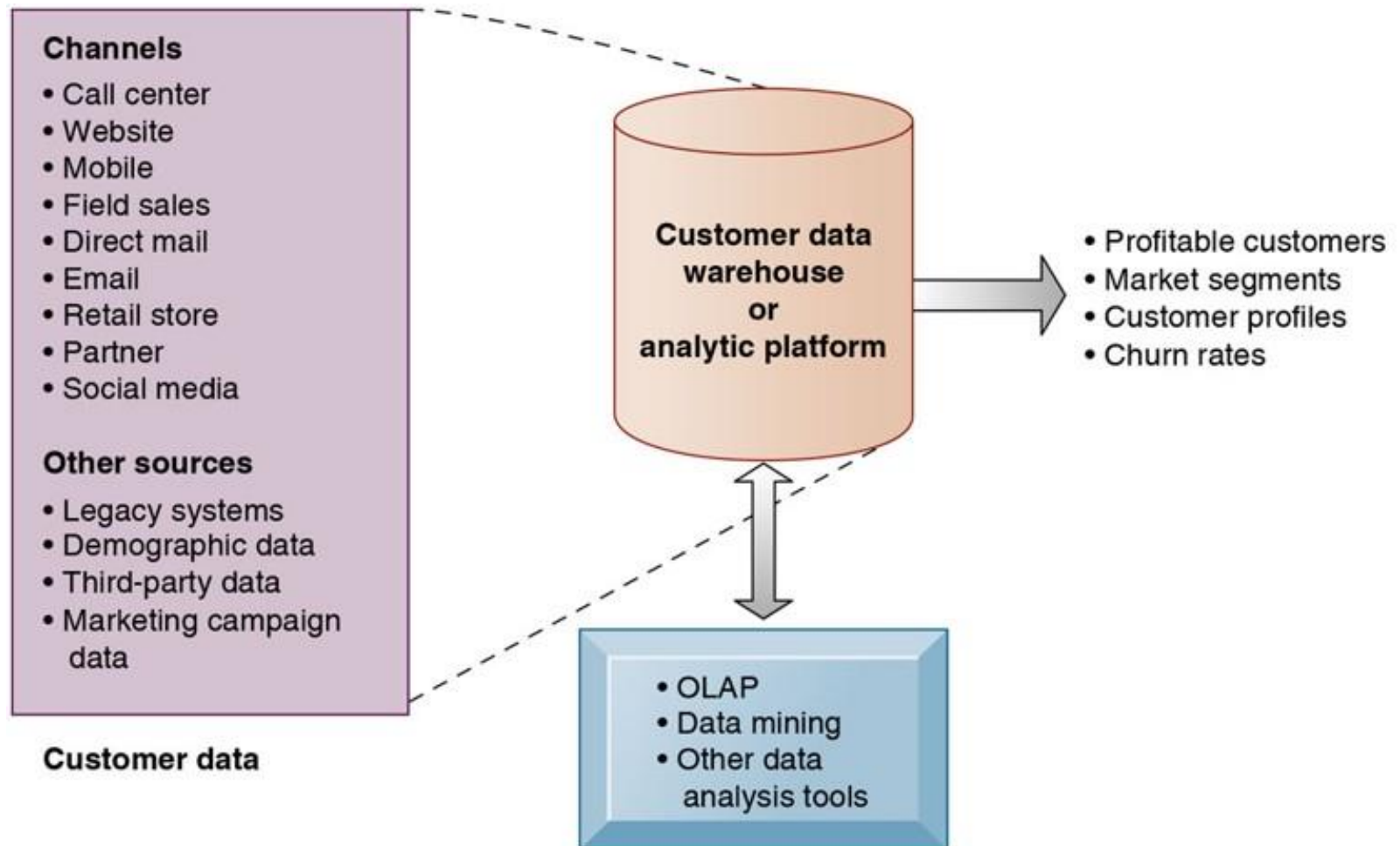


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# Operational and Analytical CRM

- Operational CRM
  - Customer-facing applications
  - Sales force automation call center and customer service support
  - Marketing automation
- Analytical CRM
  - Based on data warehouses populated by operational CRM systems and customer touch points
  - Analyzes customer data (OLAP, data mining, etc.)
    - Customer lifetime value (CLTV)

# Figure 9.10 Analytical CRM



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# Business Value of CRM Systems

- Business value of CRM systems
  - Increased customer satisfaction
  - Reduced direct-marketing costs
  - More effective marketing
  - Lower costs for customer acquisition/retention
  - Increased sales revenue
- Churn rate
  - Number of customers who stop using or purchasing products or services from a company
  - Indicator of growth or decline of firm's customer base

# CRM and AI

- Machine learning, generative AI, and other AI technologies are being integrated into CRM systems
  - To automate, enhance, and optimize CRM processes
- AI enables businesses to analyze vast amounts of customer data in real time

**Table 9.4**

**Examples of AI in CRM Systems**

| Capability                                            | Description                                                                                                                                                                                                                |
|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Digital assistant agent                               | Use AI-powered conversational bots to respond to customer inquiries by providing support, answering questions, capturing details, and resolving issues without human interaction.                                          |
| Field service routing optimization                    | Use predictive AI and machine learning with individual performance pattern data to optimize field service scheduling to maximize efficient use of resources and to help service teams complete more jobs per day.          |
| Lead conversion and sales opportunity win probability | Provide sales reps with rapid identification of their best leads and sales opportunities with predictive scoring and win probabilities. Additionally, identify sales deals that are falling behind or need more attention. |
| Pricing optimization                                  | Use AI-recommended pricing and discounts to strike a balance between achieving the most revenue with the highest likelihood to close.                                                                                      |
| Predictive scoring                                    | Use machine learning to predict which leads and acts are most likely to engage and convert to buyers.                                                                                                                      |

# Enterprise Application Challenges (1 of 3)

- Enterprise application challenges
  - Expensive to purchase and implement
    - Many projects experience cost overruns
    - Long development times
  - Technology changes
  - Business process changes
  - Organizational learning changes
  - Switching costs, dependence on software vendors
  - Data standardization, management, cleansing

# Enterprise Application Challenges (2 of 3)

- Enterprise application trends
  - Enterprise application vendors are delivering more value by becoming more
    - Flexible
    - User-friendly
    - Web-enabled
    - Mobile
    - Capable of integration with other systems
  - Cloud-based CRM systems

# Enterprise Application Challenges (3 of 3)

- Composable ERP
  - An adaptive technology strategy
  - Composable ERP
    - Brings together disparate systems and components in a way that allows them to work together as a platform

# How Enterprise Applications Are Using AI

- Enterprise application vendors have been adding business intelligence and AI tools to help managers obtain more meaningful information from the massive amounts of data these systems generate
  - Including data from the Internet of Things (IoT)
- AI-enabled tools are capable of much more rapid and complex data analysis than traditional software

**Table 9.5**

**AI Functionality Provided by Major Enterprise Application Vendors**

| Vendor     | Functions                                                                                           | AI                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SAP        | ERP, CRM, SCM, HCM                                                                                  | Machine learning and generative AI integrated with SAP business processes and solutions. For example, SAP Business AI is built into SAP finance applications to manage capital effectively, improve reporting accuracy, mitigate fraud risk and reduce losses with anomaly detection and prevention. Joule generative AI assistant acts as a work copilot across SAP business systems, answering questions with natural language.                                                                                                                                                                                               |
| Oracle     | Oracle offers a range of AI solutions, including generative AI, AI services, and AI infrastructure. | AI services, including generative AI, have prebuilt machine learning models that make it easier for developers to apply AI to applications and business operations. The models can be custom trained. Oracle Cloud Infrastructure (OCI) Generative AI is a fully managed service to seamlessly integrate these versatile language models into a wide range of use cases, including writing assistance, summarization, analysis, and chat. Oracle Fusion Cloud CX, includes generative AI-assisted answer generation, assisted scheduling for field service, opportunity quality scoring, and seller engagement recommendations. |
| Microsoft  | CRM, ERP, primarily small to medium-size companies                                                  | Microsoft Dynamics 365 AI suite includes Dynamics 365 Copilot AI assistant across all business functions. Microsoft Dynamics Customer Insights, Microsoft Dynamics Sales Insights, Microsoft Dynamics 365 Customer Service Insights are key offerings                                                                                                                                                                                                                                                                                                                                                                           |
| Salesforce | CRM                                                                                                 | Einstein suite of AI tools includes Einstein GPT Generative AI for CRM, which delivers AI-created content across every sales, service, marketing, commerce, and IT interaction.                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

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